

1.

```
GNU nano 8.7.1 address.c
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

Output:

```
vishnu@VISHNU:~/linux-memory-monitor$ nano memoryos.c
vishnu@VISHNU:~/linux-memory-monitor$ nano address.c
vishnu@VISHNU:~/linux-memory-monitor$ gcc address.c -o address
vishnu@VISHNU:~/linux-memory-monitor$ ./address
Value of x = 10
Virtual address of x = 0x7ffff70eb3594
vishnu@VISHNU:~/linux-memory-monitor$ |
```

2.

```

GNU nano 8.7.1 memory_segements.c
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

    free(heap);

    return 0;
}

```

Output:

```

vishnu@VISHNU:~/linux-memory-monitor$ nano memory_segements.c
vishnu@VISHNU:~/linux-memory-monitor$ gcc memory_segments.c -o memory_segmen
ts
cc1: fatal error: memory_segements.c: No such file or directory
compilation terminated.
vishnu@VISHNU:~/linux-memory-monitor$ gcc memory_segements.c -o memory_segem
ents
vishnu@VISHNU:~/linux-memory-monitor$ ./memory_segements
Code address : 0x5e1d037d21a9
Data address : 0x5e1d037d5010
BSS address : 0x5e1d037d5018
Heap address : 0x5e1d334f0010
Stack address : 0x7ffda9df3f7c
vishnu@VISHNU:~/linux-memory-monitor$ |

```

3.

```
vishnu@VISHNU: ~/linux-mem  ×  +  v
GNU nano 8.7.1 page_fault.c
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

Output:

```
vishnu@VISHNU:~/linux-memory-monitor$ nano page_fault.c
vishnu@VISHNU:~/linux-memory-monitor$ gcc page_fault.c -o page_fault
vishnu@VISHNU:~/linux-memory-monitor$ ./page_fault
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
vishnu@VISHNU:~/linux-memory-monitor$ |
```

4.

```
vishnu@VISHNU: ~/linux-mem  ×  +  ▾  
GNU nano 8.7.1 address_translation.c  
#include <stdio.h>  
  
int main() {  
  
    int page_table[] = {3, 5, 1, 7};  
    int page_size = 100;  
  
    int virtual_address = 250;  
  
    int page = virtual_address / page_size;  
    int offset = virtual_address % page_size;  
  
    int frame = page_table[page];  
  
    int physical_address = frame * page_size + offset;  
  
    printf("Virtual Address = %d\n", virtual_address);  
    printf("Page Number      = %d\n", page);  
    printf("Offset          = %d\n", offset);  
    printf("Frame Number     = %d\n", frame);  
    printf("Physical Address = %d\n", physical_address);  
  
    return 0;  
}
```

Output:

```
vishnu@VISHNU:~/linux-memory-monitor$ nano address_translation.c  
vishnu@VISHNU:~/linux-memory-monitor$ gcc address_translation.c -o address_translation  
cc1: fatal error: address_translation.c: No such file or directory  
compilation terminated.  
vishnu@VISHNU:~/linux-memory-monitor$ gcc address_translation.c -o address_translation  
vishnu@VISHNU:~/linux-memory-monitor$ ./address_translation  
Virtual Address = 250  
Page Number      = 2  
Offset          = 50  
Frame Number     = 1  
Physical Address = 150  
vishnu@VISHNU:~/linux-memory-monitor$ |
```

5.

```
GNU nano 8.7.1 malloc_array.c
#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);

    return 0;
}
```

Output:

```
vishnu@VISHNU:~/linux-memory-monitor$ nano malloc_array.c
vishnu@VISHNU:~/linux-memory-monitor$ gcc malloc_array.c -o malloc_array.c
gcc: fatal error: input file 'malloc_array.c' is the same as output file
compilation terminated.
vishnu@VISHNU:~/linux-memory-monitor$ gcc malloc_array.c -o malloc_array
vishnu@VISHNU:~/linux-memory-monitor$ ./malloc_array
10 20 30 40 50
vishnu@VISHNU:~/linux-memory-monitor$ |
```